

1. (Use an 8 page booklet)

[Marks 1, 2, 2, 2, 2, 3]

- (a) A coin is tossed at the same time that a die is thrown. What is the probability of getting a 3 on the die and a head on the coin?
- (b) Factor $3x^2 - x - 4$
- (c) Find the exact value of $\operatorname{cosec} 60^\circ + \tan 30^\circ$
- (d) Find integers a and b such that $(3 - \sqrt{5})^2 = a + b\sqrt{5}$
- (e) Solve $|2x - 1| = 5$
- (f) Solve the simultaneous equations: $3x - y = 5$
 $5x + 3y = -8$

2. (Use the same booklet as for question 1)

[Marks 2, 5, 5]

- (a) A die is rolled n times. Find an expression to describe the probability of rolling n sixes.
- (b) In a certain factory a machine manufactures bicycle components, of which 3% are faulty. I buy 3 such components. By drawing a tree diagram, or otherwise, find the probability that:
- (i) exactly one component is faulty.
- (ii) at least one is faulty.
- (c) A ball is dropped from a height of 3 m on to the floor. After each bounce the maximum height reached by the ball is 60% of the previous maximum height. Thus, after the first bounce it reaches a height of 180 cm, before falling again.
- (i) What is the height reached after the second bounce?
- (ii) What kind of sequence is formed by the successive heights?
- (iii) What is the total distance travelled by the ball from the time it was dropped until it comes to rest?

3. (Use a new 4 page booklet)

[Marks 2, 2, 3, 2, 1, 2]

If A, B and C are the points (5, 3), (-2, 5) and (4, -3) respectively, find the:

- (a) exact distance from A to C.
- (b) gradient of the join of A and C.
- (c) equation of the line passing through A and C, in general form.
- (d) perpendicular distance from B to the line passing through A and C.
- (e) area of the triangle ABC.
- (f) co-ordinates of the point D such that ABCD is a parallelogram.

4. (Use a new 4 page booklet)

[Marks 4, 3, 5]

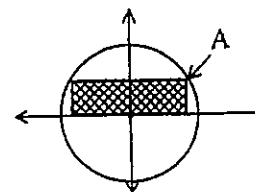
- (a) For the equation $x^2 + (m-3)x + m = 0$ find the values of m for which the equation has
- (i) two equal roots
- (ii) two distinct real roots
- (b) Find all real roots of the equation $4^x - 6 \times 2^x + 8 = 0$
- (c) For the parabola $2y^2 - 4x - 8 = 0$, find
- (i) the co-ordinates of the vertex
- (ii) the co-ordinates of the focus
- (iii) the equation of the directrix

5. (Use a new 8 page booklet)

[Marks 2, 4, 6]

- (a) Differentiate $\log_e \left(\frac{2x-3}{x+1} \right)$ with respect to x leaving your answer without simplifying it.
- (b) Show that the graph of $y = e^x \cos x$ has a stationary point at $x = \frac{\pi}{4}$ and determine its nature.

(c)

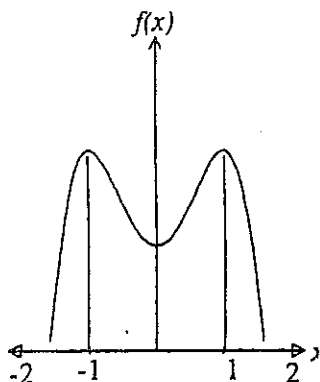
The diagram shows a rectangle inscribed in a semicircle whose equation is $y = \sqrt{4 - x^2}$.

- (i) Let the x coordinate of point A be x . Show the area of the rectangle is given by $2x\sqrt{4 - x^2}$.
- (ii) Find the maximum area of the rectangle.

6. (Use the same booklet as for question 5)

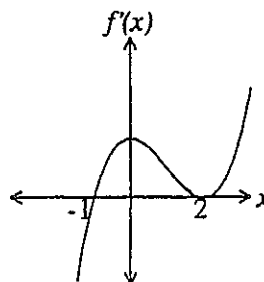
[Marks 5, 4, 3]

- (a) The graph shows part of the function $y = 2\cos(1-x^2)$. Using 5 function values, find an approximation using the trapezoidal rule for the area between this graph, the x -axis and the lines $x = 1$ and $x = -1$.



- (b) The area enclosed by the graph of $y = \frac{1}{\sqrt{1-x}}$, the coordinate axes and the line $x = \frac{1}{2}$ is rotated around the x axis. Calculate the volume of the solid so formed.

- (c) The diagram below shows the graph of the derived function $f'(x)$, for the function $y = f(x)$.



- (i) Explain the behaviour of the graph of $y = f(x)$ at $x = -1$ and at $x = 2$.
 (ii) Explain what you know about the graph of $y = f(x)$ when x is between these values.

7. (Use a new 8 page booklet)

[Marks 2, 2, 2, 4, 2]

- (a) Find the derivatives, without simplifying, of each of:

(i) $e^{\cos 3x}$ (ii) $\sqrt{xe^{2x}}$ (iii) $\tan^3(5x+4)$

- (b) (i) Sketch, on the same set of axes, graphs of $y = 3\sin 2x$ and $y = \frac{x}{2}$
 (ii) State the number of solutions of the equation $6\sin 2x - x = 0$

8. (Use the same booklet as for question 7)

[Marks 3, 3, 3, 3]

- (a) Show that the graph of $y = \log_e x$ is concave down and increasing for all values of $x > 0$
 (b) Show that if $y = A \cos 4x + B \sin 4x$ then $\frac{d^2 y}{dx^2} + 16y = 0$.
 (c) The area under the curve $y = \frac{1}{x}$ between $x = 1$ and $x = b$ is equal to 1 square unit. What is the value of b ?
 (d) If the gradient of the tangent at the point (x, y) on a particular curve is $2\sin 3x$ and the curve passes through the point $(\frac{\pi}{3}, \frac{8}{3})$, find the equation of the curve.

9. (Use a new 8 page booklet)

[Marks 1, 2, 2, 3, 4]

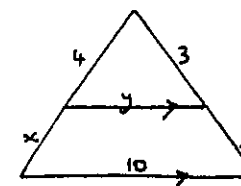
- (a) Given $F(x) = \frac{x+1}{x-1}$, $G(x) = x^2$, $H(x) = \sqrt{x}$

- (i) State the domain of $F(x)$.
 (ii) Evaluate, where possible, $HG(-3)$ and $GH(-3)$.
 [If either is not possible, explain why.]

- (iii) Show that $F\left(\frac{1}{x}\right) = -F(x)$.

- (iv) Show that $F^{-1}(x) = F(x)$.

- (b) From the diagram evaluate x and y .



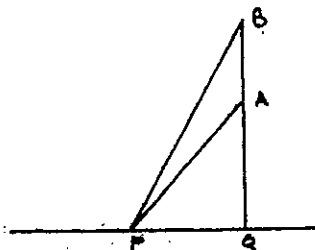
10. (Use the same booklet as for question 9)

[Marks 4, 3, 5]

- (a) A point $P(x, y)$ moves so that its distance from $(-1, 1)$ is equal to its distance from the straight line $3x - 4y + 4 = 0$.

Show that the cartesian equation of the locus is $16x^2 + 26x + 24xy + 9y^2 - 18y + 34 = 0$.

- (b) AB is a 12 m flag pole on the top of a building. Find the length of BP given that $\angle QPB = 56^\circ 37'$ and $\angle QPA = 43^\circ 19'$.



- (c) ABCD is a parallelogram. J and K are points of trisection of the diagonal AC (i.e. $AJ = JK = KC$).

(i) Draw a clear diagram to represent this situation.

(ii) Prove that $\triangle ADJ \cong \triangle CBK$.

(iii) Hence explain why $DJ \parallel BK$.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} \quad (n \neq -1; x \neq 0 \text{ if } n < 0)$$

$$\int \frac{1}{x} dx = \log_e x \quad (x > 0)$$

$$\int \cos ax dx = \frac{1}{a} \sin ax \quad (a \neq 0)$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax \quad (a \neq 0)$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} \quad (a \neq 0)$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} \quad (a > 0, -a < x < a)$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \log_e \left\{ x + \sqrt{x^2 - a^2} \right\} \quad (|x| > |a|)$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \log_e \left\{ x + \sqrt{x^2 + a^2} \right\}$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} \quad (a \neq 0)$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax \quad (a \neq 0)$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax \quad (a \neq 0)$$

Question 1

(a) $P(3) = \frac{1}{6}$ $P(H) = \frac{1}{2}$

$P(3 \text{ AND } H) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$

(b) Factorize $3x^2 - x - 4 = (3x - 4)(x + 1)$

$3x - 4$
 $x + 1$

(c) $\csc 60^\circ + \tan 30^\circ = \frac{1}{\sin 60^\circ} + \tan 30^\circ$

$= \frac{1}{\frac{\sqrt{3}}{2}} + \frac{1}{\sqrt{3}}$

$= \frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}}$

$= \frac{3}{\sqrt{3}}$

$= \sqrt{3}$

$= \boxed{\sqrt{3}}$

(d) $(3 - \sqrt{5})^2 = (3)^2 - 2(3)(\sqrt{5}) + (\sqrt{5})^2$

$= 9 - 6\sqrt{5} + 5$

$= 14 - 6\sqrt{5}$

$\therefore a = 14$ $b = -6$

(e) $|2x - 1| = 5$

$2x - 1 = 5$ OR $-(2x - 1) = 5$

$2x = 6$ $2x - 1 = -5$

$x = 3$ $2x = -4$

$x = -2$

Check Solutions

$\therefore x = 3$ AND $x = -2$ ARE Solutions.

(f) Solving

$3x - y = 5 \dots (1)$

$5x + 3y = -8 \dots (2)$

Elimination

$(1) \times 3$ $9x - 3y = 15 \dots (3)$

$5x + 3y = -8 \dots (2) +$

$14x = 7$

$\therefore x = \frac{1}{2}$

Sub $x = \frac{1}{2}$ into eqn (1) to find y

ie $3(\frac{1}{2}) - y = 5$

$\frac{3}{2} - y = 5$

$y = -5 + \frac{3}{2}$

$y = \frac{-10 + 3}{2}$

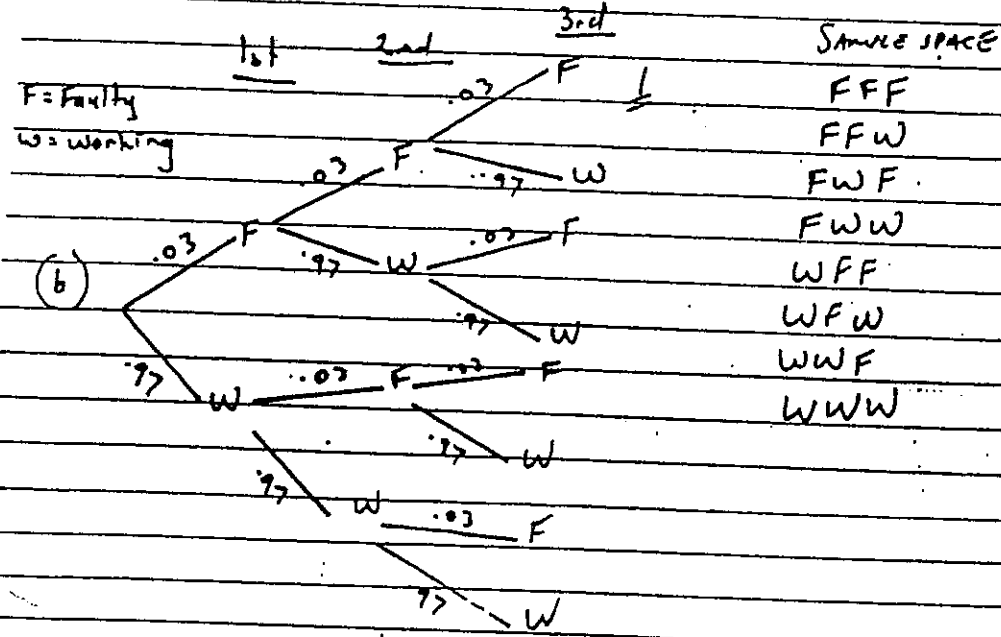
$\therefore y = -\frac{7}{2}$

Hence Solution to Simult equations

is $(\frac{1}{2}, -\frac{7}{2})$

Question 2

(a) $(\frac{1}{6})^n$



(i) $P(\text{EXACTLY ONE Faulty}) = P(FWW) + P(WFW) + P(WWF)$

$= (0.03 \times 0.97 \times 0.97) + (0.97 \times 0.03 \times 0.97) + (0.97 \times 0.97 \times 0.03)$

$= (0.028) + (0.028) + (0.028)$

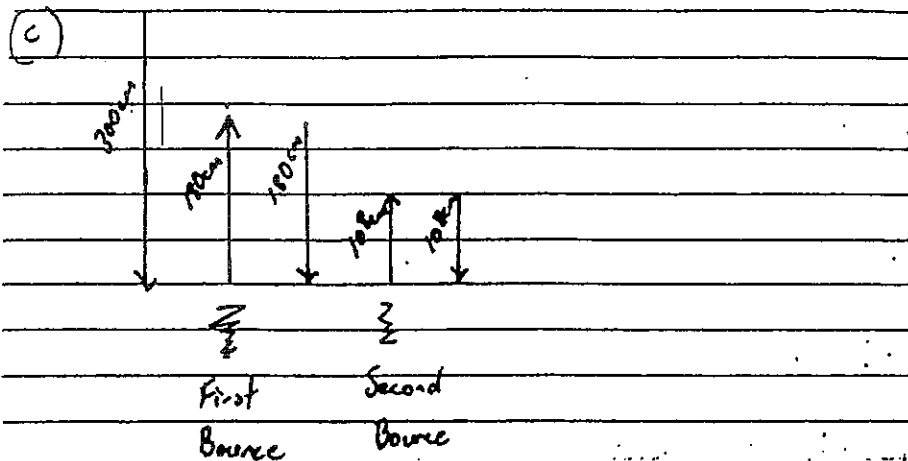
$= 0.084$

(ii) $P(\text{at least one Faulty}) = 1 - P(WWW)$

$= 1 - (0.97 \times 0.97 \times 0.97)$

$= 1 - 0.913$

$= 0.087$



(i) $1.8m$ or $108cm$ $\frac{1}{2}$

(iii) accept Infinite Sequence or Geometric or limiting height

(iii) Total Distance = $3 + 2(1.8 + 1.08 + 0.648)$ $\frac{1}{2}$

$$= 3 + 2 \left(\frac{a}{1-r} \right)$$

$$= 3 + 2 \left(\frac{1.8}{1-0.6} \right) \frac{1}{2}$$

$$= 3 + 9$$

$\approx$

So Ball travels = $12m$ $\frac{1}{2}$ or $1200cm$

Q3. $A(5,3)$, $B(-2,5)$, $C(4,-3)$

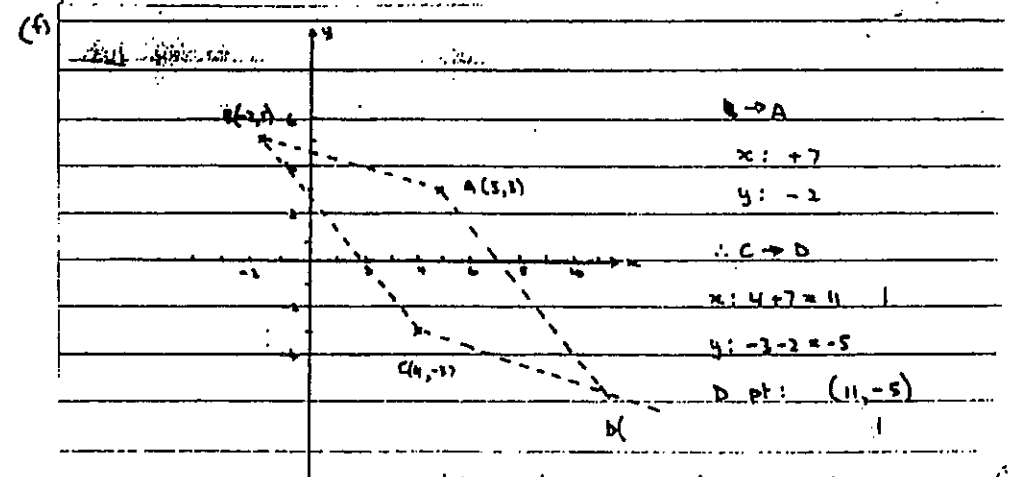
(a) $d_{AC} = \sqrt{(5-4)^2 + (3-(-3))^2}$
 $= \sqrt{1+36} = \sqrt{37}$

(b) $m_{AC} = \frac{6}{1} = 6$

(c) $y-3 = 6(x-5)$ $|$ $y-3 = 6x-30$ $|$
 $6x - y - 27 = 0$

(d) $d = \frac{|6(-2) - 5 - 27|}{\sqrt{6^2 + 1^2}} = \frac{44}{\sqrt{37}}$

(e) $A = \frac{1}{2} \cdot \frac{44}{\sqrt{37}} \cdot \sqrt{37} = 22 \text{ u}^2$



Note: $(-3, -1)$ gives a point in the parallelogram ABCD. You were asked for ABCD.

4. (a) $\Delta = (n-3)^2 - 4m$
 $= n^2 - 10n + 9$

(i) Two equal roots if $n^2 - 10n + 9 = 0$
 $(n-9)(n-1) = 0$
 $n = 1, 9$

(ii) Two distinct roots if $n^2 - 10n + 9 > 0$
 $\therefore n < 1 \text{ or } n > 9$

(b) $4^x - 6 \times 2^x + 8 = 0$
 $(2^x - 2)(2^x - 4) = 0$
 $\therefore 2^x = 2, 4$
 $x = 1, 2$

(c) $2y^2 - 5x - 8 = 0$
 $y^2 = 2(x+2)$
 (i) Vertices = $(-2, 0)$
 (ii) Focus = $(-1, 0)$
 (iii) Directrix $x = -3$

131

QUESTION 5

(a) Outcomes assessed: H3, P7

Marking Criteria

Criteria	Marks
1 mark for correct differentiation procedure	2
1 mark for correct answer	

Answer

$$y = \ln\left(\frac{2x-3}{x+1}\right)$$

$$= \ln(2x-3) - \ln(x+1)$$

$$\therefore \frac{dy}{dx} = \frac{2}{2x-3} - \frac{1}{x+1} \quad \text{OR} \quad \frac{x+1}{2x-3} \times \frac{2(x-1) - (2x+3)}{(x+1)^2}$$

(b) Outcomes assessed: H5, P6, P7

Criteria	Marks
1 mark for derivative	4
1 mark for showing derivative 0 at $x = \frac{\pi}{4}$	
1 mark for either sign diag for derivative or second derivative	
1 mark for correct conclusion from the above	

Answer

$$y = e^x \cos x$$

$$y' = e^x \cos x - e^x \sin x$$

$$\text{at } x = \frac{\pi}{4}, y' = e^{\frac{\pi}{4}} \left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right) = 0$$

$$\therefore \text{stationary pt at } x = \frac{\pi}{4}$$

$$y'' = e^x (\cos x - \sin x) + e^x (-\sin x - \cos x) = -2e^x \sin x$$

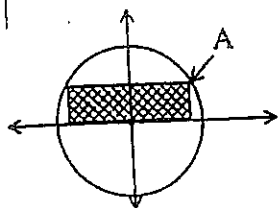
$$\text{at } x = \frac{\pi}{4}, y'' = -2e^{\frac{\pi}{4}} \times \frac{1}{\sqrt{2}} < 0$$

$$\therefore \text{maximum turning point}$$

(c) Outcomes assessed: H1, H2, H5

Criteria	Marks
1 mark for identifying dimensions of rectangle	6
1 mark for area	
1 mark for derivative of area	
1 mark for solution of $a' = 0$	
1 mark for justifying maximum	
1 mark for max area = 4	

Answer:



A's coordinates are $(x, \sqrt{4-x^2})$

Thus dimensions of rectangle are $2x \times \sqrt{4-x^2}$

ie, area = $2x\sqrt{4-x^2}$

$$a = 2x\sqrt{4-x^2}$$

$$\frac{da}{dx} = 2 \times \sqrt{4-x^2} + 2x \times \frac{1}{\sqrt{4-x^2}} \times -2x$$

$$= \frac{8-4x^2}{\sqrt{4-x^2}}$$

$$= 0 \text{ if } x^2 = 2$$

$$x = \sqrt{2} \text{ (must be positive)}$$

$$\frac{da}{dx} > 0 \text{ if } x < \sqrt{2} \text{ and } \frac{da}{dx} < 0 \text{ if } x > \sqrt{2}$$

ie, max turning point

$$\text{and } a_{\max} = 2\sqrt{2} \times \sqrt{2} = 4$$

QUESTION 6

(a) Outcomes assessed: P3, P4, H5,

Criteria	Marks
2 marks for correct function values 2 marks for substituting correctly into a correct formula 1 mark for evaluation from correct formula	5

Answer:

x	-1	-0.5	0	0.5	1
y	2	1.463...	1.081...	1.463...	2

$$\text{area} = \frac{0.5}{2} (2 + 2 + 2(1.463 + 1.081 + 1.463))$$

$$= 3.0036...$$

$$= 3.00 \text{ units of area (2d.p.)}$$

(b) Outcomes assessed: P3, H8

Criteria	Marks
1 mark for correct volume formula 1 mark for integral of correct x function 1 mark for correct integral 1 mark for evaluation	4

Answer

$$V = \pi \int_0^{\frac{1}{2}} y^2 dx$$

$$= \pi \int_0^{\frac{1}{2}} \frac{1}{1-x} dx$$

$$= \pi [-\ln(1-x)]_0^{\frac{1}{2}}$$

$$= \pi \ln 2$$

(c) Outcomes assessed: P7, P8, H2, H5, H6, H7

Criteria	Marks
1 mark for identifying the minimum at $x = -1$ 1 mark for identifying the stationary inflexion at $x = 2$ OR 1 mark for identifying that there is a stationary point at each of $x = -1$ and $x = 2$ 1 mark for identifying the inflexion between $x = -1$ and $x = 2$ OR identifying that $f(x)$ is increasing between these values	3

Answer:

(i)

$$\text{at } x = -1 \quad f'(x) = 0$$

$$f'(-1^-) < 0$$

$$f'(-1^+) > 0$$

$\therefore$ minimum turning point

$$\text{at } x = 2, \quad f'(x) = 0$$

$$f'(2^+) > 0 \text{ and } f'(2^-) > 0$$

$\therefore$ stationary inflexion

(ii)

$$\text{for } -1 < x < 2 \quad f'(x) > 0$$

$\therefore f(x)$ is increasing

also, in this interval the gradient of $f'(x)$ changes from positive to negative, so there

is a of cur ie, an inflexion at $x = 2$

QUESTION 7

(a) (i)

Outcomes assessed: P7 H3

Marking Criteria

Criteria	Marks
• 1 mark for correct differentiation procedure	2
• 1 mark for correct answer	

Answer

$$\frac{d}{dx} e^{\cos 3x} = e^{\cos 3x} \times -\sin 3x \times 3$$

(ii) Outcomes assessed: P7

Marking Criteria

Criteria	Marks
• 1 mark for correct differentiation procedure	2
• 1 mark for correct answer	

Answer

$$\begin{aligned} \text{(iv)} \quad \frac{d}{dx} \sqrt{x} e^x &= x^{\frac{1}{2}} (e^x)^{\frac{1}{2}} \quad \text{or} \quad \frac{d}{dx} (x e^{2x})^{\frac{1}{2}} \\ &= \frac{1}{2} (x e^{2x})^{-\frac{1}{2}} \times [x e^{2x} + e^{2x}] \\ &= \frac{1}{2} \cdot \frac{1}{\sqrt{x}} \cdot \frac{1}{e} \times e^{2x} (2x+1) \\ &= e^x \left(\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{x}} \right) \\ &= e^x \left(\frac{1}{\sqrt{x}} + \frac{1}{2\sqrt{x}} \right) \approx \text{before} \end{aligned}$$

(iii) Outcomes assessed: P7

Marking Criteria

Criteria	Marks
• 1 mark for correct differentiation procedure	2
• 1 mark for correct answer	

Answer

$$\begin{aligned} \frac{d}{dx} \tan^3(5x+4) &= \frac{d}{dx} [\tan(5x+4)]^3 \\ &= 3 [\tan(5x+4)]^2 \cdot \sec^2(5x+4) \cdot 5 \\ &= 15 \tan^2(5x+4) \sec^2(5x+4) \end{aligned}$$

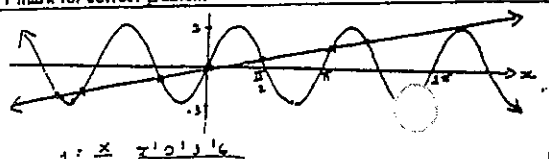
(b)

(i) Outcomes assessed: P4 H5

Marking Criteria

Criteria	Marks
• 1 mark for sine graph	4
• 1 mark for correct range & period	
• 1 mark for straight line graph	
• 1 mark for correct gradient	

Answer



Criteria	Marks
• 1 mark for linking given equation to the graph	2
• 1 mark for correct answer according to graph	

Answer (i) $6 \sin 2x - x = 0$
 $\text{ie } 6 \sin 2x = x$
 $\text{ie } 3 \sin 2x = \frac{x}{2}$
 Since $y = 3 \sin 2x$ and $y = \frac{x}{2}$ intersect 7 times
 $6 \sin 2x - x = 0$ has 7 solutions

QUESTION 8

(a) Outcomes assessed: P5 H6

Marking Criteria

Criteria	Marks
• 1 mark for 1st and 2nd derivatives	3
• 1 mark for correct link to increasing curve	
• 1 mark for correct link to concave down curve	

(b) Outcomes assessed: P7 H5

Criteria	Marks
• 1 mark for 1st derivative	3
• 1 mark for 2nd derivative	
• 1 mark for proof	

Answers

(a) $\frac{dy}{dx} = \frac{1}{x}$

$\frac{d^2y}{dx^2} = -\frac{1}{x^2}$

When $x > 0$, $\frac{1}{x} > 0 \therefore$ INCREASING

$-\frac{1}{x^2} < 0 \therefore$ CONCAVE DOWN

Hence $y = \log_e x$ is concave down + increasing for all $x > 0$

(c) Outcomes assessed: P8 H8

Criteria	Marks
• 1 mark for identifying the integral	3
• 1 mark for integration	
• 1 mark for substitution to correct answer	

(d) Outcomes assessed: P8 H5

Criteria	Marks
• 1 mark for integration	3
• 1 mark for substitution	
• 1 mark for correct answer	

Answers

(c) $\int_1^b \frac{1}{x} dx = 1$ ✓

$[\log_e x]_1^b = 1$

$\log_e b - \log_e 1 = 1$

$\log_e b - 0 = 1$

$\therefore b = e$

OR $\int_b^1 \frac{1}{x} dx = 1$

$[\log_e x]_b^1 = 1$

$\log_e 1 - \log_e b = 1$

$-\log_e b = 1$

$\therefore b = e^{-1}$

(d) $\frac{dy}{dx} = 2 \sin 3x$

$y = -\frac{2}{3} \cos 3x + C$ ✓

$\frac{\pi}{2} = -\frac{2}{3} \cos(3 \cdot \frac{\pi}{2}) + C$

$\frac{\pi}{2} = -\frac{2}{3} \cos(\frac{3\pi}{2}) + C$

$\frac{\pi}{2} = -\frac{2}{3} \cdot 1 + C$

$C = \frac{\pi}{2} + \frac{2}{3}$

$\therefore y = -\frac{2}{3} \cos 3x + \frac{\pi}{2} + \frac{2}{3}$ ✓

(1)

7(a) $F(x) = \frac{x+1}{x-1}$, $G(x) = x^2$, $H(x) = \sqrt{x}$ (i) Domain $F(x)$ $\forall x \in \mathbb{R}$ except $x=1$

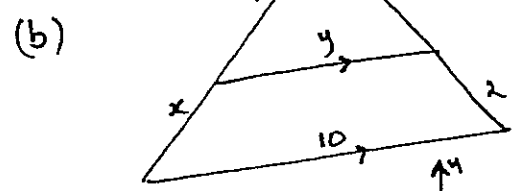
(ii) $H(G(-3)) = H(9) = 3$. $G(H(-3)) = G(\sqrt{-3})$ But $\sqrt{-3}$ is not a Real Number $\therefore$ Value $\sqrt{-3}$ not useable. $\therefore G(\sqrt{-3})$ not possible. (1)

(iii) $F(\frac{1}{x}) = \frac{\frac{1}{x} + 1}{\frac{1}{x} - 1}$
 $= \frac{1+x}{1-x}$ or $-\frac{(1+x)}{x-1}$ $G - F(x) = -\frac{(1+x)}{x-1}$
 $\therefore F(\frac{1}{x}) = -F(x)$ (2)

(iv) For inverse of $F(x)$ rewrite as $y = \frac{x+1}{x-1}$ & for inverse (x, y) is $x = \frac{y+1}{y-1}$

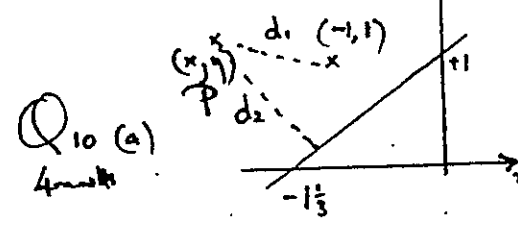
$\therefore x(y-1) = y+1$ & $xy - y = x+1$
 So $y(x-1) = x+1$ or $y = \frac{x+1}{x-1}$ But this y is $f^{-1}(x)$ (3)

$\therefore F^{-1}(x) = \frac{x+1}{x-1}$



From the diagram & considering similar Δ s
 $\frac{4}{x} = \frac{3}{y} \Rightarrow x = \frac{8}{3}$ or $2\frac{2}{3}$ (2)
 & $\frac{y}{10} = \frac{3}{5} \Rightarrow y = 6$. (2)

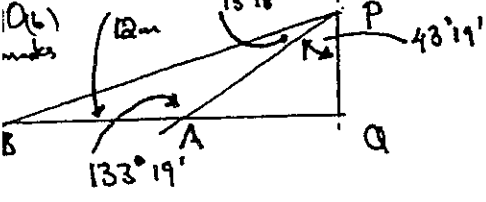
134



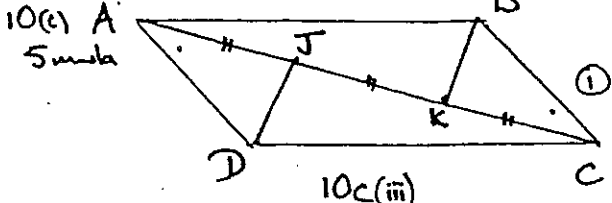
Let d_1 be distance from P to $(-1, 1)$ is $d_1 = \sqrt{(x+1)^2 + (y-1)^2}$
 & let d_2 be distance P to line is $d_2 = \frac{|3x - 4y + 4|}{\sqrt{9+16}}$ (1)

But $d_1 = d_2 \Rightarrow d_1^2 = d_2^2 \Rightarrow (x+1)^2 + (y-1)^2 = \frac{(3x - 4y + 4)^2}{25}$ (1)

$25(x^2 + 2x + 1) + 25(y^2 - 2y + 1) = (3x - 4y + 4)^2$
 $25x^2 + 50x + 25 + 25y^2 - 50y + 25 = 9x^2 + 24x + 16y^2 - 32y - 24xy + 16$
 So the required equation of the locus is $16x^2 + 26x + 24xy + 9y^2 - 18y + 34 = 0$ (1)



From the diagram
 $\frac{BP}{\sin 133^\circ 19'} = \frac{12}{\sin 13^\circ 16'}$ (2)
 $\therefore BP = 37.95213536$
 ie $BP = 37.95m$ (1)



10(c)(ii) Consider ΔADJ & ΔCBK
 $\angle ADJ = \angle CBK$ given
 $\angle DAJ = \angle BCK$ alt $\angle$ $AD \parallel BC$
 $AD = BC$ opp sides of parallelogram
 $\therefore \Delta ADJ \equiv \Delta CBK$ SAS
 $\therefore \angle AJD = \angle CKB$ Corresponding $\angle$ s Congruent Δ (1)

Since $\angle AJD = \angle CKB$
 then $\angle DJK = \angle BKT$
 Adj $\angle$ s on st line
 $\therefore DJ \parallel BK$
 Alternate $\angle$ s equal.